

The flattening bijection

Why the double sum $\sum_a \sum_\mu$ and a single sum $\sum_p$ compute the same drive in a multi-sequence Kuramoto DAM

1 The two forms

For a network of N phase oscillators storing K sequences, where sequence a consists of P_a patterns $\xi_i^{a,\mu} \in \{\pm 1\}^N$, the multi-sequence dynamics are

$$\dot{\theta}_i = \omega_i - \sin \theta_i \sum_{a=1}^K \sum_{\mu=1}^{P_a} \xi_i^{a,\mu+1} \left(\frac{1}{N-1} \sum_{j \neq i} \xi_j^{a,\mu} \cos \theta_j \right)^d. \quad (1)$$

The claim to be justified is that the double sum in (1) may be replaced by a *single* sum over a flat list of $P_{\text{tot}} = \sum_a P_a$ patterns,

$$\dot{\theta}_i = \omega_i - \sin \theta_i \sum_{p=1}^{P_{\text{tot}}} \Xi_i^{p,\text{next}} \left(\frac{1}{N-1} \sum_{j \neq i} \Xi_j^p \cos \theta_j \right)^d, \quad (2)$$

for suitably defined flattened arrays Ξ^p and $\Xi^{p,\text{next}}$, and that this replacement is *exact* — not an approximation, not a limit, and not contingent on any property of the stored patterns.

2 The index set and the summand

Everything outside the double sum in (1) — the term ω_i and the prefactor $-\sin \theta_i$ — is independent of a and μ , so it suffices to consider the *drive*

$$V_i := \sum_{a=1}^K \sum_{\mu=1}^{P_a} \xi_i^{a,\mu+1} \left(\frac{1}{N-1} \sum_{j \neq i} \xi_j^{a,\mu} \cos \theta_j \right)^d.$$

Two objects deserve names. First the *index set*

$$I := \{ (a, \mu) : 1 \leq a \leq K, 1 \leq \mu \leq P_a \}, \quad |I| = \sum_{a=1}^K P_a = P_{\text{tot}},$$

and second the *summand*, a function of a pattern χ and its designated successor χ' (with θ and i held fixed):

$$F_i(\chi, \chi') := \chi'_i \left(\frac{1}{N-1} \sum_{j \neq i} \chi_j \cos \theta_j \right)^d.$$

With this notation the drive is simply a sum of $|I|$ real numbers indexed by I :

$$V_i = \sum_{(a,\mu) \in I} F_i(\xi^{a,\mu}, \xi^{a,\mu+1}). \quad (3)$$

Note that I is a set of *pairs*; nothing about writing the sum as two nested symbols $\sum_a \sum_\mu$ is essential. The nesting is a way of enumerating I , and the question is whether a different enumeration gives the same total.

3 The bijection

Definition 1 (Offsets). Let $O_1 := 0$ and $O_{a+1} := O_a + P_a$ for $1 \leq a \leq K$, so that $O_a = \sum_{b < a} P_b$ and $O_{K+1} = P_{\text{tot}}$.

Definition 2 (Flattening map). Define $\varphi : I \rightarrow \{1, \dots, P_{\text{tot}}\}$ by

$$\varphi(a, \mu) := O_a + \mu.$$

This is exactly “concatenate the sequences in order”: sequence a ’s patterns occupy the consecutive block of positions $\{O_a + 1, \dots, O_{a+1}\}$.

Lemma 1. φ is a bijection.

Proof. Well-defined. For $(a, \mu) \in I$ we have $1 \leq \mu \leq P_a$, hence $O_a + 1 \leq \varphi(a, \mu) \leq O_a + P_a = O_{a+1}$. Since $O_1 = 0$ and $O_{K+1} = P_{\text{tot}}$, the value lies in $\{1, \dots, P_{\text{tot}}\}$.

The blocks partition the codomain. The intervals $B_a := \{O_a + 1, \dots, O_{a+1}\}$, for $a = 1, \dots, K$, are consecutive and non-overlapping, and their union is $\{1, \dots, P_{\text{tot}}\}$. (Immediate from $O_1 = 0$, $O_{K+1} = P_{\text{tot}}$ and $O_a < O_{a+1}$ whenever $P_a \geq 1$.)

Injective. Suppose $\varphi(a, \mu) = \varphi(a', \mu')$. The common value lies in exactly one block B_a , so $a = a'$; then $O_a + \mu = O_a + \mu'$ gives $\mu = \mu'$.

Surjective. Given $p \in \{1, \dots, P_{\text{tot}}\}$, let a be the unique index with $p \in B_a$, i.e. $O_a < p \leq O_{a+1}$, and set $\mu := p - O_a$. Then $1 \leq \mu \leq P_a$, so $(a, \mu) \in I$ and $\varphi(a, \mu) = p$. $\square$

We write $\varphi^{-1}(p) = (a(p), \mu(p))$ for the inverse, with $a(p)$ the unique a such that $O_a < p \leq O_{a+1}$, and $\mu(p) = p - O_{a(p)}$.

Remark (Uniform lengths). When every sequence stores the same number of patterns, $P_a \equiv P$, the offsets are $O_a = (a - 1)P$ and the map collapses to

$$\varphi(a, \mu) = (a - 1)P + \mu,$$

which is precisely row-major flattening of a $K \times P$ grid. In array terms, if the patterns are held as a tensor of shape (K, P, N) , then φ is the reshape to shape (KP, N) ; the offsets never need to be materialised, because the block structure is recoverable from the shape alone.

4 Invariance of finite sums under reindexing

Proposition 1 (Reindexing). Let I be a finite set, $f : I \rightarrow \mathbb{R}$ any function, and $\varphi : I \rightarrow J$ a bijection. Then

$$\sum_{\iota \in I} f(\iota) = \sum_{j \in J} f(\varphi^{-1}(j)).$$

Proof. Induction on $|I|$. For $|I| = 0$ both sides are the empty sum. Suppose the claim holds for all sets of size n and let $|I| = n + 1$. Pick any $\iota_0 \in I$ and put $j_0 = \varphi(\iota_0)$. Then φ restricts to a bijection $I \setminus \{\iota_0\} \rightarrow J \setminus \{j_0\}$, and by generalised associativity and commutativity of addition in $\mathbb{R}$,

$$\sum_{\iota \in I} f(\iota) = f(\iota_0) + \sum_{\iota \in I \setminus \{\iota_0\}} f(\iota) = f(\iota_0) + \sum_{j \in J \setminus \{j_0\}} f(\varphi^{-1}(j)) = \sum_{j \in J} f(\varphi^{-1}(j)). \quad \square$$

The content of Proposition 1 is worth stating plainly, because it is the entire justification for flattening: **a finite sum depends only on the multiset of its terms, not on how they are enumerated.** The hypotheses are that the index set is finite and that φ is a bijection. There are *no* hypotheses on f or on the values it takes.

5 The derivation

Define the flattened arrays by transporting the pattern and its successor along φ^{-1} :

$$\Xi^p := \xi^{a(p), \mu(p)}, \quad \Xi^{p, \text{next}} := \xi^{a(p), \mu(p)+1}. \quad (4)$$

Applying Proposition 1 to (3) with the bijection φ of Lemma 1 and $f(a, \mu) = F_i(\xi^{a, \mu}, \xi^{a, \mu+1})$:

$$V_i = \sum_{(a, \mu) \in I} F_i(\xi^{a, \mu}, \xi^{a, \mu+1}) = \sum_{p=1}^{P_{\text{tot}}} F_i(\Xi^p, \Xi^{p, \text{next}}) = \sum_{p=1}^{P_{\text{tot}}} \Xi_i^{p, \text{next}} \left(\frac{1}{N-1} \sum_{j \neq i} \Xi_j^p \cos \theta_j \right)^d,$$

which is the drive appearing in (2). Since the prefactor $-\sin \theta_i$ and the term ω_i were untouched, (1) and (2) define the same vector field, for every $\theta \in \mathbb{R}^N$ and every choice of stored patterns. $\blacksquare$

6 The hypothesis that can fail

The derivation used the bijection only to *relabel* terms. What makes the result non-vacuous in practice is that the flattened arrays must genuinely satisfy (4). The first of these, $\Xi^p = \xi^{a(p), \mu(p)}$, holds by construction of any concatenation. The second is where an implementation can go wrong.

Let σ_a denote sequence a 's successor map on its own pattern indices,

$$\sigma_a(\mu) = \mu + 1 \quad (\mu < P_a), \quad \sigma_a(P_a) = \begin{cases} P_a & \text{self-referencing boundary } (\xi^{a, P_a+1} := \xi^{a, P_a}), \\ 1 & \text{cyclic boundary } (\xi^{a, P_a+1} := \xi^{a, 1}). \end{cases}$$

Then (4) says the flattened successor index must be

$$\Sigma(p) = \varphi(a(p), \sigma_{a(p)}(\mu(p))), \quad (5)$$

that is: *decode to (a, μ) , advance within sequence a , re-encode*.

Failure mode. It is tempting to define the flattened successor as $\Sigma(p) = p + 1$ — “shift the concatenated array down by one row”. This agrees with (5) only when p lies in the interior of a block, and is wrong at every block boundary.

Take $K = 2$ with $P_1 = 3$ and $P_2 = 2$, so $P_{\text{tot}} = 5$ and $O = (0, 3, 5)$, under the self-referencing boundary:

p	1	2	3	4	5
$(a(p), \mu(p))$	(1, 1)	(1, 2)	(1, 3)	(2, 1)	(2, 2)
$\Xi^{p, \text{next}}$ (correct, via (5))	$\xi^{1,2}$	$\xi^{1,3}$	$\xi^{1,3}$	$\xi^{2,2}$	$\xi^{2,2}$
Ξ^{p+1} (naive global shift)	$\xi^{1,2}$	$\xi^{1,3}$	$\xi^{2,1}$	$\xi^{2,2}$	undefined

At $p = 3$ the naive shift makes the last pattern of sequence 1 drive towards the *first pattern of sequence 2* — a cross-sequence transition that appears nowhere in (1), and which would couple two sequences that the model treats as independent chains. At $p = 5$ it is undefined (or wraps to $\xi^{1,1}$, equally wrong).

The remedy is structural rather than arithmetical: resolve each sequence's successor array *before* concatenating, so that the boundary condition is applied along that sequence's own index range. Concatenating K already-correct successor arrays yields (5) automatically, with no offset arithmetic at the boundaries. Stated abstractly, the requirement is that φ intertwine the successor structure: the flattened successor map must be the conjugate $\varphi \circ \sigma \circ \varphi^{-1}$ of the per-sequence one, where σ acts on I by $\sigma(a, \mu) = (a, \sigma_a(\mu))$.

7 Two remarks on scope

Exactness in $\mathbb{R}$ versus bitwise agreement in floating point

Proposition 1 is a statement about $\mathbb{R}$, where addition is associative. IEEE-754 floating-point addition is commutative but *not* associative, so two enumerations of the same terms are guaranteed to agree as real numbers yet may differ in their last bits once rounded.

The identity therefore guarantees mathematical equality; it does *not* by itself guarantee that two implementations produce bit-identical output. What does guarantee that is the additional observation that φ of Definition 2 is *order-preserving* with respect to the lexicographic order on I : enumerating I as $(1, 1), (1, 2), \dots, (K, P_K)$ and enumerating $\{1, \dots, P_{\text{tot}}\}$ in increasing order visit the terms in the same order, so the two accumulations perform the identical sequence of roundings. A flattening that permuted the blocks — still a perfectly valid bijection, still exact in $\mathbb{R}$ — would in general give a different floating-point result.

What the argument does not establish

The reindexing identity is narrow, and it is worth being explicit about its limits. It says only that two ways of organising the *same* finite collection of terms produce the same number. It says nothing whatsoever about:

- whether cueing near one stored pattern retrieves the intended sequence;
- how large the interference from the $K - 1$ other sequences is, or how it scales;
- how many sequences can be stored before retrieval degrades (capacity);
- whether the terminal pattern is an attractor, or whether a cyclic boundary sustains a limit cycle.

Each of those is a question about the *values* of the terms and the resulting flow, and would require its own analysis or measurement. The flattening argument is prior to all of them: it establishes that the two ways of writing the model are the same model, and nothing more.